AI · Multi-Agent

Technical Report · May 2026

Job Agent

AI-Powered Personal Job Search Assistant

Autonomous discovery · Intelligent matching · LLM-tailored resumes

6 AGENTS · LANGGRAPH ORCHESTRATION DAG

PLATFORMS	SCORING	COST / APPLY	DAILY JOBS	LLM ROUTING
5	6D	$0.07	400+	2+1
scraped daily	match dimensions	resume + cover letter	collected & deduplicated	DeepSeek + Claude Opus

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Multi-Agent · LangGraph · Python 3.12 · FastAPI · Apify

v0.1.0 · May 2026

Active Development

§ Executive Summary

EXECUTIVE SUMMARY

Job Agent orchestrates six specialized AI agents to automate the full job search pipeline — daily discovery across five platforms, 6-dimension LLM scoring, and two-round resume tailoring — for under $0.75/day. DeepSeek handles high-volume analytical tasks; Claude Opus is reserved for quality-critical prose. Every application decision remains human-controlled.

6	<$1	100%
Collaborating AI Agents	Daily pipeline cost	Human-controlled
LangGraph orchestration	Full 5-platform run	No auto-submission

1 Problem Statement

The modern job search presents a core tension: candidates must evaluate hundreds of listings daily to find genuine fits, yet each application demands a carefully tailored resume. Existing tools either fully automate submission (sacrificing quality) or offer only passive aggregation (shifting the burden back to the user).

🔴 Current Pain Points

- Manual search across 5+ platforms daily — repetitive
- No systematic multi-dimension job-fit evaluation
- Generic resumes reduce interview conversion rates
- Cross-platform duplicate listings waste review time
- No centralized application status tracking
- Comparable SaaS tools cost $200–500/month

🟢 What Job Agent Solves

- Daily auto-collection: LinkedIn, Indeed, Dice, YC, HN
- Structured 6D LLM scoring for every collected job
- Two-round AI tailoring: gap analysis + resume rewrite
- 3-layer deduplication removes cross-platform duplicates
- SQLite tracking with full history and version control
- Full active campaign under $35/month

Explicit Non-Goal: Job Agent does *not* auto-submit applications. Every submission is reviewed and submitted manually — prioritizing quality over volume.

2 Goals & Objectives

G1 Automate Job Discovery
Daily collection across all major platforms with preference-based pre-filtering before LLM processing.

G2 Objective Multi-Dimensional Scoring
6-dimension structured LLM scoring producing a calibrated 0–100 score with transparent reasoning.

G3 High-Quality Resume Tailoring at Low Cost
Two-round, two-model pipeline delivering ATS-optimized resumes at under $0.10 per application.

G4 Centralized Application Tracking
Persistent, queryable SQLite database of all jobs, scores, artifacts, and history — accessible via CLI and Web UI.

G5 Cost-Efficient Operation
Route LLM tasks by cost-quality fit: DeepSeek for volume, Claude Opus for quality-critical tasks. Target: under $35/month.

G6 Production-Ready Infrastructure
Liveness, readiness, and per-agent health endpoints compatible with Kubernetes and systemd. Cron-schedulable.

3 System Architecture

The system is a **LangGraph DAG** of six specialized agents communicating via shared typed state. Each agent wraps discrete CLI components and can be invoked standalone or as part of the full automated pipeline.

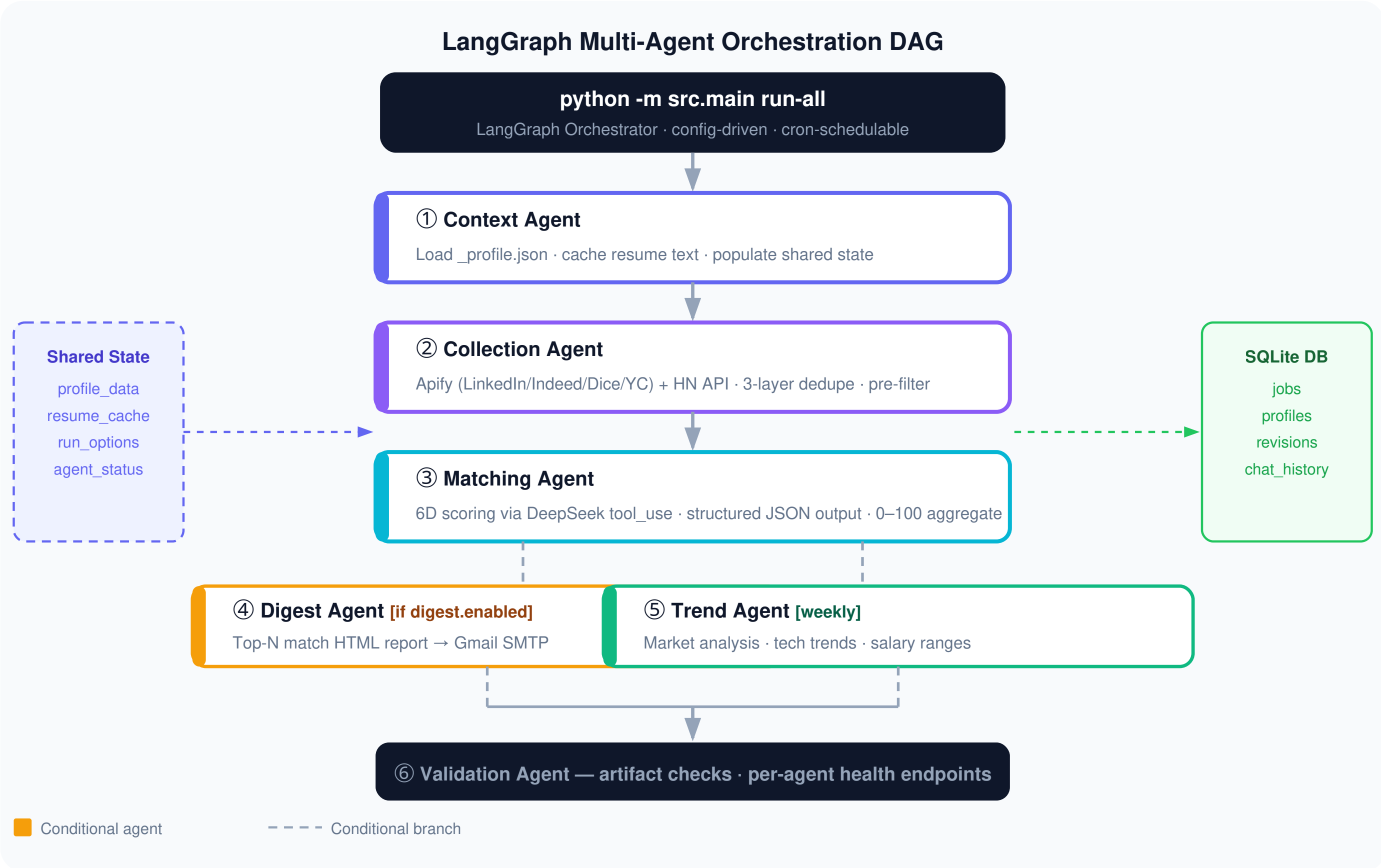

Figure 1 — LangGraph DAG. Six agents share typed state; SQLite persists all records.

Technology Stack

Layer	Technology	Purpose
Orchestration	LangGraph (Python 3.12)	Agent DAG, shared state, conditional branching
LLM — Analytical	DeepSeek Chat	Scoring, planning, profile analysis, trends
LLM — Quality	Claude Opus 4	Resume execution, cover letter generation
Scraping	Apify Actors + Playwright	Managed anti-bot scraping across platforms
Web Server	FastAPI	Job UI, resume refinement chat, health probes
Database	SQLite / SQLAlchemy	Jobs, scores, application history, chat sessions
Scheduler	cron / systemd	Daily collection, weekly trend reports

4 Job Collection & Deduplication

All platforms share a `BaseCollector` interface. LinkedIn/Indeed use Apify-managed actors to bypass anti-bot measures; HackerNews uses the free Firebase + Algolia API directly.

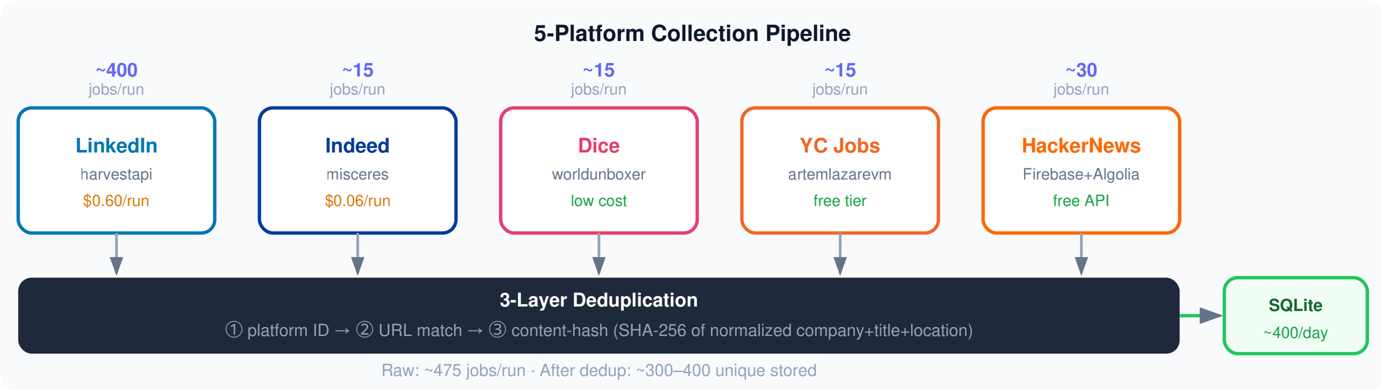

Figure 2 — 5-platform collection pipeline with per-run volumes, costs, and 3-layer deduplication.

5 6-Dimension Match Scoring

DeepSeek evaluates each job via `tool_use` structured output, scoring each dimension independently before computing a 0–100 aggregate plus reusable downstream artifacts.

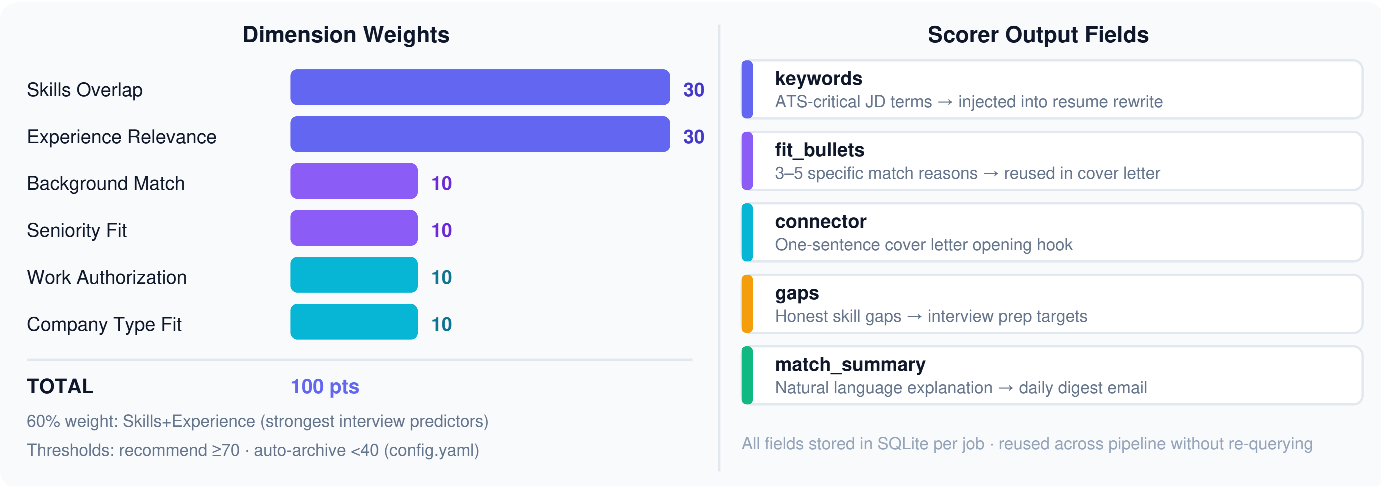

Figure 3 — Scoring weights (left) and structured output fields reused across the pipeline (right).

6 Resume Tailoring Pipeline

A two-round, two-model approach separates cheap gap analysis from quality prose generation — achieving frontier quality at sub-$0.10 per application.

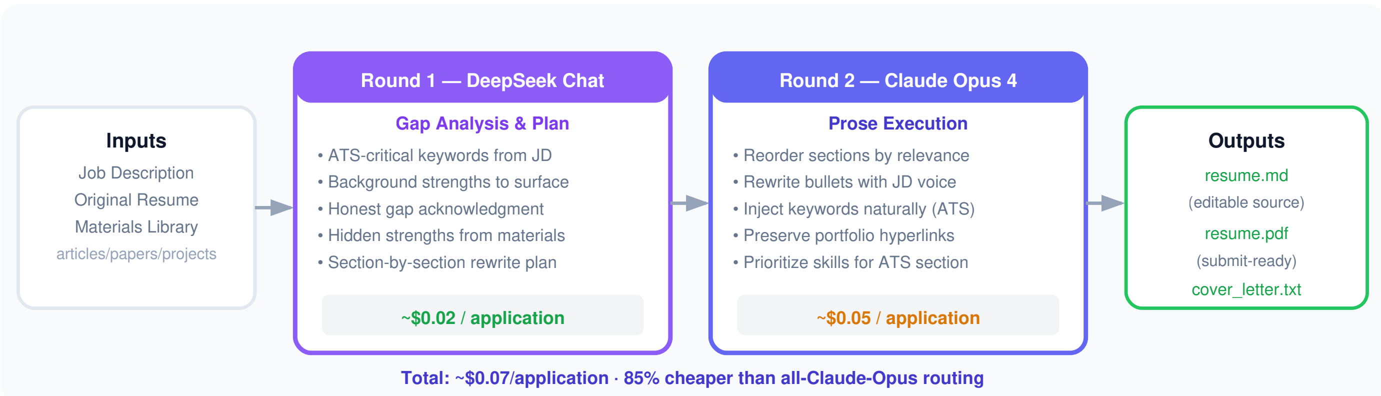

Figure 4 — Two-round tailoring pipeline. DeepSeek for analysis; Claude Opus for quality prose execution.

7 LLM Model Routing Strategy

`agent.py` routes tasks to DeepSeek or Claude Opus via a unified Anthropic SDK client. DeepSeek exposes an Anthropic-compatible endpoint — provider switching is a config-file change, no code changes required.

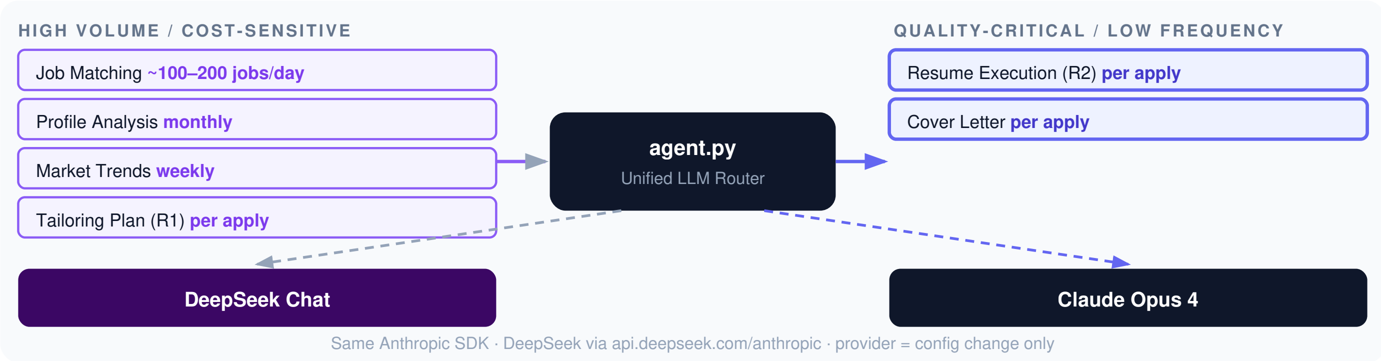

Figure 5 — LLM routing. 85% of daily volume routes to DeepSeek; Claude Opus reserved for prose quality.

8 Cost Analysis

Figure 6 — Monthly cost composition. Scraping dominates; LLM routing keeps compute negligible.

Component	Model / Service	Frequency	Unit Cost	Monthly
LinkedIn collection	Apify harvestapi	Daily	$0.60/run	$18
Indeed collection	Apify misceres	Daily	$0.06/run	$1.80
Dice + YC + HN	Apify / Free API	Daily	<$0.05/run	~$1.50
DeepSeek matching	deepseek-chat	Daily	~$0.03/run	~$0.90
Resume + cover letter	Claude Opus 4	Per apply	~$0.07/job	~$0.35 (5 apps)
Monthly Total (30 days + 5 applications)				**~$22–35**

9 Key Design Decisions

① Apify over Direct Playwright

LinkedIn and Indeed's Cloudflare Bot Fight Mode made direct scraping brittle. Apify manages anti-bot evasion in managed infrastructure. At $0.60/day for LinkedIn, the cost is justified by reliability and maintenance savings.

② SQLite over PostgreSQL

Zero-configuration portability for single-user use. The serial pipeline (collect → match → digest) means SQLite's limited concurrent-write support is not a constraint. PostgreSQL migration path exists for multi-user scale.

③ Prompts as Files, not Code

All LLM instructions in `prompts/*.md` — reviewable in version control, iterable without code changes, and reloadable at runtime. Enables non-developer adjustment of model behavior.

④ No Auto-Submission

Auto-apply generates high-volume, low-quality submissions that damage candidate reputation. Job Agent invests compute in one excellent tailored artifact per application — quality over quantity.

⑤ Materials Library Integration

`data/materials/` (articles, papers, projects) is injected into the tailoring context, surfacing evidence beyond the resume — e.g., citing a published eBPF article when applying to kernel-adjacent roles.

⑥ 3-Round Profile Analysis

A 3-round LLM dialogue (Extraction → Multi-Perspective → Top-10 Synthesis) generates AI-extracted target roles with aliases, broader terms, and market demand scores — replacing a static keyword list.

10 Conclusion & Roadmap

Job Agent demonstrates that purposeful multi-agent LLM design can automate the most time-consuming aspects of job search at sub-dollar daily cost while preserving human judgment. The core insight is **task routing**: DeepSeek for volume, Claude Opus for quality — reducing daily LLM cost by ~85%.

> **Core Outcome:** Full active campaign (5 platforms, 400+ jobs/day, 5 tailored applications/month) for ~$22–35/month vs. $200–500/month for comparable SaaS tools.

Priority	Feature	Description
High	Greenhouse / Lever ATS APIs	Direct ATS integration for improved reliability and data quality
High	A/B prompt experimentation	Track which tailoring variants produce higher interview conversion rates
Medium	Glassdoor salary integration	Enrich job records with salary data for compensation benchmarking
Medium	Interview prep agent	Role-specific question lists from the 6D match output and gaps field
Low	PostgreSQL backend	Multi-user support for shared or team installations